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Cationic lipids and transfection methods

Abstract

The present invention relates in part to novel cationic lipids and their use, e.g., in delivering nucleic acids to cells.

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Background/Summary

PRIORITY (1) The present application is a continuation of U.S. application Ser. No. 17/553,564, filed Dec. 16, 2021, which is a continuation of Ser. No. 16/930,901, filed Jul. 16, 2020 (now U.S. Pat. No. 11,242,311), which is a continuation of U.S. application Ser. No. 16/746,279, filed Jan. 17, 2020 (now U.S. Pat. No. 10,752,576), which is a continuation of U.S. application Ser. No. 16/660,317, filed Oct. 22, 2019 (now U.S. Pat. No. 10,611,722), which is a continuation of U.S. application Ser. No. 16/526,621, filed Jul. 30, 2019 (now U.S. Pat. No. 10,501,404), the content of each of which is hereby incorporated herein by reference in its entirety.

FIELD OF THE INVENTION
(1) The present invention relates, in part, to various novel lipids, including methods, compositions, and products for delivering nucleic acids to cells.
BACKGROUND
(2) Lipid-based materials, such as liposomes, are used as biological carriers for pharmaceutical and other biological applications, e.g., to introduce agents into cultured cell lines. Lipids are commonly used to deliver nucleic acids to cells in vitro under low-serum or serum-free conditions, for instance in transfection. However, serum components inhibit the activity of many lipids, limiting their use in the presence of serum, both in vitro and in vivo.
(3) Improved lipid delivery systems, e.g., to achieve higher levels of transfection both in vitro and in vivo, are desirable. In particular, lipid delivery systems that are active in the presence of serum are needed. Improved levels of transfection will allow the treatment of disease states for which higher levels of expression than are currently achievable with lipid delivery systems are needed for therapeutic effect. Alternatively, higher transfection levels will allow for use of smaller amounts of material to achieve comparable expression levels, thereby decreasing potential toxicities and decreasing cost.
(4) There is a need for novel lipids, lipid-like materials, and lipid-based delivery systems in the art.
SUMMARY OF THE INVENTION
(5) Accordingly, the present invention relates to new lipids that find use, inter alia, in improved delivery of biological payloads, e.g. nucleic acids, to cells.
(6) In aspects, the present invention relates to a compound of Formula (I)
(7) ##STR00001##
where n is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15.
(8) In embodiments, n is 1-12, or 2-12, or 1-10, or 2-10, or 1-8, or 2-8, or 2-6.
(9) In embodiments, the present invention relates to compound (i):

(10) ##STR00002##
(11) In embodiments, the present invention relates to compound (ii):
(12) ##STR00003##
(13) In embodiments, the present invention relates to compound (iii):
(14) ##STR00004##
(15) In embodiments, the present invention relates to compound (iv):
(16) ##STR00005##
(17) In embodiments, the present invention relates to compound (v):
(18) ##STR00006##
(19) In embodiments, the present invention relates to compound (vi):
(20) ##STR00007##
(21) In embodiments, the present invention relates to compound (vii):
(22) ##STR00008##
(23) In embodiments, the present invention relates to compound (viii):
(24) ##STR00009##
(25) In embodiments, the present compounds (e.g. of Formula I) are components of a pharmaceutical composition and/or a lipid aggregate and/or a lipid carrier and/or a lipid nucleic-acid complex and/or a liposome and/or a lipid nanoparticle.
(26) In embodiments, the present compounds (e.g. of Formula I) are components of a pharmaceutical composition and/or a lipid aggregate and/or a lipid carrier and/or a lipid nucleic-acid complex and/or a liposome and/or a lipid nanoparticle which does not require an additional or helper lipid.
(27) In embodiments, the present compounds (e.g. of Formula I) are components of a pharmaceutical composition and/or a lipid aggregate and/or a lipid carrier and/or a lipid nucleic-acid complex and/or a liposome and/or a lipid nanoparticle which comprises a nucleic acid, such as DNA (e.g., without limitation, a plasmid, cosmid, phage, recombinant virus or other vector) or RNA (e.g., without limitation, an siRNA, micro-RNA (miRNA), long non-coding RNA (lncRNA), an in vitro transcribed RNA, a synthetic RNA, and/or an mRNA, in each case that comprises one or more non-canonical nucleotides that confer stability, avoid degradation by one or more nucleases, and/or avoid substantial cellular toxicity, or does not comprise a non-canonical nucleotide).
(28) In aspects, the present invention relates to a method for transfecting a cell with a nucleic acid, comprising contacting the cell with a complex of the nucleic acid and a compound described herein (e.g. of Formula I), where the complex of the nucleic acid and the compound described herein (e.g. of Formula I) is optionally formed prior to contact with the cell.
(29) In embodiments, the transfection method provides at least one of the following characteristics: (a) high transfection efficiency, (b) high level of endosomal escape, (c) serum-resistance, (d) low toxicity effects, (e) high level of protein expression, (f) transfectability in various cell types, and (g) transfectability without additional lipids or reagents for transfection, e.g. relative to a method of transfecting a cell with complex of the nucleic acid and DOTMA, DODMA, DOTAP, DODAP, DOPE, cholesterol, LIPOFECTIN (cationic liposome formulation), LIPOFECTAMINE (cationic liposome formulation), LIPOFECTAMINE 2000 (cationic liposome formulation), LIPOFECTAMINE 3000 (cationic liposome formulation), and combinations thereof.
(30) The details of the invention are set forth in the accompanying description below. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, illustrative methods and materials are now described. Other features, objects, and advantages of the invention will be apparent from the description and from the claims. In the specification and the appended claims, the singular forms also include the plural unless the context clearly dictates otherwise. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs.

(31) Any aspect or embodiment disclosed herein can be combined with any other aspect or embodiment as disclosed herein.

Description

DETAILED DESCRIPTION OF THE FIGURES

(1) FIG. **1** depicts primary human epidermal keratinocytes cultured in a 24-well plate, and transfected with 400 ng per well of in vitro transcribed RNA encoding green fluorescent protein (GFP) complexed with the indicated lipid and with the indicated mass ratio of lipid to RNA. Complexation was performed in DMEM, and transfections were performed in 100% fetal bovine serum (FBS). Images were taken eight hours following transfection.

(2) FIG. **2** depicts the experiment of FIG. **1**, with fluorescence measured at the indicated time points following transfection using DHDLinS.

(3) FIG. **3** depicts the results of an experiment conducted as in FIG. **1**, but with the indicated amounts of RNA (in nanograms) and the indicated lipid-to-RNA mass ratios (in micrograms of lipid per microgram of RNA). Images were taken 16 hours following transfection. As shown in the figure, all RNA amounts and lipid-to-RNA mass ratios tested yielded a fluorescent signal.

(4) FIG. **4** depicts the results of an experiment conducted as in FIG. **1**, but with human peripheral blood mononuclear cells (hPBMCs) instead of keratinocytes. Images were taken 16 hours following transfection. “LF3000” indicates cells transfected with LIPOFECTAMINE 3000 (cationic liposome formulation) commercial transfection reagent. “Neg.” indicates un-transfected cells.

(5) FIG. **5**A depicts the results of an experiment conducted as in FIG. **4**, but with a confluent layer of primary human epidermal keratinocytes instead of hPBMCs. Images were taken 24 hours following transfection.

(6) FIG. **5**B depicts the experiment of FIG. **5**A, shown at higher magnification.

(7) FIG. **6**A depicts the results of an experiment conducted as in FIG. **4**, but with primary human adult dermal fibroblasts instead of hPBMCs. Images were taken 16 hours following transfection.

(8) FIG. **6**B depicts the experiment of FIG. **6**A, shown at higher magnification.

(9) FIG. **7** depicts the results of an experiment conducted as in FIG. **3**, but with DHDLinS purified by extraction with acetone as described in Example 5.

(10) FIG. **8** depicts primary human epidermal keratinocytes cultured in a 24-well plate, and transfected with 100 ng per well of in vitro transcribed RNA encoding green fluorescent protein (GFP) complexed with the compounds of Formula I, where n is as indicated. Images were taken eight hours following transfection.

(11) FIG. **9**A depicts the measured 500 MHz proton NMR spectrum of DHDLinS in deuterated chloroform.

(12) FIG. **9**B depicts DHDLinS.

DETAILED DESCRIPTION OF THE INVENTION

(13) The present invention is based, in part, on the discovery of novel lipids that, inter alia, demonstrate superior abilities to support delivery of nucleic acids to cells, e.g. during transfection. The present invention provides such compositions, methods of making the compositions, and methods of using the compositions to introduce nucleic acids into cells, including for the treatment of diseases.

(14) Compounds

(15) In aspects, the present invention relates to a compound of Formula (I)

(16) ##STR00010##

where n is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15.

(17) In embodiments, n is 1-14, or 1-12, or 1-10, or 1-8, or 1-6, or 1-4, or 1-2, or 2-14, or 2-12, or

2-10, or 2-8, or 2-6, or 2-4, or 4-14, or 4-12, or 4-10, or 4-8, or 4-6, or 6-14, or 6-12, or 6-10, or 6-8, or 8-14, or 8-12, or 8-10, or 10-14, or 10-12.
(18) In embodiments, n is 1. In embodiments, n is 2. In embodiments, n is 3. In embodiments, n is 4. In embodiments, n is 5. In embodiments, n is 6. In embodiments, n is 7. In embodiments, n is 8. In embodiments, n is 9. In embodiments, n is 10. In embodiments, n is 11. In embodiments, n is 12. In embodiments, n is 13. In embodiments, n is 14. In embodiments, n is 15.
(19) In embodiments, the present invention relates to compound (i):
(20) ##STR00011##
(21) In embodiments, the present invention relates to compound (ii):
(22) ##STR00012##
(23) In embodiments, the present invention relates to compound (iii):
(24) ##STR00013##
(25) In embodiments, the present invention relates to compound (iv):
(26) ##STR00014##
(27) In embodiments, the present invention relates to compound (v):
(28) ##STR00015##
(29) In embodiments, the present invention relates to compound (vi):
(30) ##STR00016##
(31) In embodiments, the present invention relates to compound (vii):
(32) ##STR00017##
(33) In embodiments, the present invention relates to compound (viii):
(34) ##STR00018##
(35) In embodiments, the present invention relates to a pharmaceutical composition and/or a lipid aggregate and/or a lipid carrier and/or a lipid nucleic-acid complex and/or a liposome and/or a lipid nanoparticle which comprises a compound described herein (e.g. of Formula I).
(36) In embodiments, the pharmaceutical composition and/or lipid aggregate and/or lipid carrier and/or lipid nucleic-acid complex and/or liposome and/or lipid nanoparticle is in any physical form including, e. g., lipid nanoparticles, liposomes, micelles, interleaved bilayers, etc.
(37) In embodiments, the pharmaceutical composition and/or lipid aggregate and/or lipid carrier is a liposome. In embodiments, the liposome is a large unilamellar vesicle (LUV), multilamellar vesicle (MLV) or small unilamellar vesicle (SUV). In embodiments, the liposome has a diameter up to about 50 to 80 nm. In embodiments, the liposome has a diameter of greater than about 80 to 1000 nm, or larger. In embodiments, the liposome has a diameter of about 50 to 1000 nm, e.g. about 200 nm or less. Size indicates the size (diameter) of the particles formed. Size distribution may be determined using quasi-elastic light scattering (QELS) on a Nicomp Model 370 sub-micron particle sizer.
(38) In embodiments, the compound (e.g. of Formula I), and/or pharmaceutical composition and/or lipid aggregate and/or lipid carrier and/or lipid nucleic-acid complex and/or liposome and/or lipid nanoparticle comprising the compound (e.g. of Formula I), is soluble in an alcohol (e.g. ethyl alcohol) at room temperature (e.g. about 20-25° C.) and/or at low temperatures (e.g. about 0° C., or about −10° C., or about −20° C., or about −30° C., or about −40° C., or about −50° C., or about −60° C., or about −70° C., or about −80° C.).
(39) In certain embodiments, the present invention relates to methods and compositions for producing lipid-encapsulated nucleic acid particles in which nucleic acids are encapsulated within a lipid layer. Such nucleic acid-lipid particles, including, without limitation incorporating RNAs, can be characterized using a variety of biophysical parameters including: drug to lipid ratio; encapsulation efficiency; and particle size. High drug to lipid ratios, high encapsulation efficiency, good nuclease resistance and serum stability and controllable particle size, generally less than 200 nm in diameter can, in certain situations, be desirable (without limitation).
(40) Nucleic acid to lipid ratio can refer to the amount of nucleic acid in a defined volume of

preparation divided by the amount of lipid in the same volume. This may be on a mole per mole basis, or on a weight per weight basis, or on a weight per mole basis, or on a mole per weight basis. For final, administration-ready formulations, the nucleic acid to lipid ratio may be calculated after dialysis, chromatography and/or enzyme (e.g., nuclease) digestion has been employed to remove as much external nucleic acid as possible.

(41) Encapsulation efficiency can refer to the drug (including nucleic acid) to lipid ratio of the starting mixture divided by the drug (including nucleic acid) to lipid ratio of the final, administration competent formulation. This can be a measure of relative efficiency. For a measure of absolute efficiency, the total amount of nucleic acid added to the starting mixture that ends up in the administration competent formulation, can also be calculated. The amount of lipid lost during the formulation process may also be calculated. Efficiency can be used as a measure of the wastage and expense of the formulation.

(42) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles have utility for delivery of macromolecules and other compounds into cells. In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles have utility for delivery of nucleic acids into cells.

(43) In embodiments, there is provided a method for transfecting a cell with a nucleic acid, comprising contacting the cell with a complex of the nucleic acid and a present compound (e.g. of Formula I) and/or pharmaceutical composition and/or lipid aggregate and/or lipid carrier and/or lipid nucleic-acid complex and/or liposome and/or lipid nanoparticle. In embodiments, the complex of the nucleic acid and the present compound (e.g. of Formula I) and/or pharmaceutical composition and/or lipid aggregate and/or lipid carrier and/or lipid nucleic-acid complex and/or liposome and/or lipid nanoparticle is formed prior to contact with the cell.

(44) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles encapsulate nucleic acids with high-efficiency, and/or have high drug to lipid ratios, and/or protect the encapsulated nucleic acid from degradation and/or clearance in serum, and/or are suitable for systemic delivery, and/or provide intracellular delivery of the encapsulated nucleic acid. In addition, in embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles are well-tolerated and provide an adequate therapeutic index, such that patient treatment at an effective dose of the nucleic acid is not associated with significant toxicity and/or unacceptable risk to the patient.

(45) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles are polycationic. In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles form stable complexes with various anionic macromolecules, such as polyanions, such as nucleic acids, such as RNA or DNA. In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles, have the property, when dispersed in water, of forming lipid aggregates which strongly, via their cationic portion, associate with polyanions. In embodiments, by modulating the amount of cationic charges relative to the anionic compound, for example by using an excess of cationic charges relative to the anionic compound, the polyanion-lipid complexes may be adsorbed on cell membranes, thereby facilitating uptake of the desired compound by the cells.

(46) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles mediate one or more of (i) compacting a nucleic acid payload to be delivered, which may protect it from nuclease degradation and/or may enhance receptor-mediated uptake, (ii) improving association with negatively-charged cellular membranes, which may be modulated by giving the complexes a positive charge, (iii) promoting fusion with endosomal membranes, which may facilitate the release of complexes from endosomal compartments, and (iv) enhancing transport from the cytoplasm to the nucleus.

(47) In embodiments, the present invention relates to the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles for transfection, or methods of transfection, which have a high transfection efficiency. In embodiments, the transfection efficiency is measured by assaying a percentage of cells that are transfected compared to the entire population, during a transfection protocol. In various embodiments, the transfection efficiency of the present compositions and methods is greater than about 30%, or greater than about 40%, or greater than about 50%, or greater than about 60%, or greater than about 70%, or greater than about 80%, or greater than about 90%, or greater than about 95%. In various embodiments, the transfection efficiency of the present compositions and methods is greater than the transfection efficiency of commercially available products (e.g. LIPOFECTIN (cationic liposome formulation), LIPOFECTAMINE (cationic liposome formulation), LIPOFECTAMINE 2000 (cationic liposome formulation), LIPOFECTAMINE 3000 (cationic liposome formulation) (Life Technologies)). In various embodiments, the transfection efficiency of the present compositions and methods is about 1.1-fold, or about 1.5-fold, or about 2-fold, or about 5-fold, or about 10-fold, or about 15-fold, or about 20-fold, or about 30-fold, or about 50-fold, or greater than about 50-fold greater than the transfection efficiency of commercially available products (e.g. LIPOFECTIN (cationic liposome formulation), LIPOFECTAMINE (cationic liposome formulation), LIPOFECTAMINE 2000 (cationic liposome formulation), LIPOFECTAMINE 3000 (cationic liposome formulation) (Life Technologies)).

(48) In embodiments, the present invention relates to the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers for transfection, or methods of transfection, which permit a high level of endosomal escape. In various embodiments, the endosomal escape of the present compositions and methods is greater than the endosomal escape of commercially available products (e.g. LIPOFECTIN (cationic liposome formulation), LIPOFECTAMINE (cationic liposome formulation), LIPOFECTAMINE 2000 (cationic liposome formulation), LIPOFECTAMINE 3000 (cationic liposome formulation) (Life Technologies)). In various embodiments, the endosomal escape of the present compositions and methods is about 5-fold, or 10-fold, or 15-fold, or 20-fold, or 30-fold greater than the endosomal escape of commercially available products (e.g. LIPOFECTIN (cationic liposome formulation), LIPOFECTAMINE (cationic liposome formulation), LIPOFECTAMINE 2000 (cationic liposome formulation), LIPOFECTAMINE 3000 (cationic liposome formulation) (Life Technologies)).

(49) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles are serum-resistant. In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles are substantially stable in serum. In embodiments, the present transfection methods can function in the presence of serum and/or do not require serum inactivation and/or media changes. In embodiments, the stability in serum and/or serum-resistance is measurable via in vitro assays known in the art. Transfection efficiency in varying amounts of serum may be used to assess the ability to transfect a macromolecule (e.g., without limitation, DNA or RNA), optionally in comparison to commercially

available products (e.g. LIPOFECTIN (cationic liposome formulation), LIPOFECTAMINE (cationic liposome formulation), LIPOFECTAMINE 2000 (cationic liposome formulation), LIPOFECTAMINE 3000 (cationic liposome formulation) (Life Technologies)).

(50) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles for transfection, or methods of transfection, have low or reduced toxicity effects. In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles for transfection, or methods of transfection, have reduced toxicity effects as compared to commercially available products (e.g. LIPOFECTIN (cationic liposome formulation), LIPOFECTAMINE (cationic liposome formulation), LIPOFECTAMINE 2000 (cationic liposome formulation), LIPOFECTAMINE 3000 (cationic liposome formulation) (Life Technologies)). In various embodiments, the present compositions and methods allow for cells having greater than about 50%, or about 60%, or about 70%, or about 80%, or about 90%, or about 95% viability after transfection. In various embodiments, the present compositions and methods allow for cells having about 1.1-fold, or about 1.5-fold, or about 2-fold, or about 5-fold, or about 10-fold, or about 15-fold, or about 20-fold, or about 30-fold greater viability after transfection, as compared to commercially available products (e.g. LIPOFECTIN (cationic liposome formulation), LIPOFECTAMINE (cationic liposome formulation), LIPOFECTAMINE 2000 (cationic liposome formulation), LIPOFECTAMINE 3000 (cationic liposome formulation) (Life Technologies)). In embodiments, toxicity effects include disruption of cell morphology and/or viability or deregulation of one or more genes.

(51) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles for transfection, or methods of transfection, permit a high level of protein expression from the nucleic acid (e.g. DNA or RNA) being transfected. In various embodiments, the protein expression of the present compositions and methods is greater than about 30%, or greater than about 40%, or greater than about 50%, or greater than about 60%, or greater than about 70%, or greater than about 80%, or greater than about 90%, or greater than about 95% more than in un-transfected cells and/or cells contacted with naked nucleic acid. In various embodiments, the resultant protein expression of the present compositions and methods is greater than the resultant protein expression of commercially available products (e.g. LIPOFECTIN (cationic liposome formulation), LIPOFECTAMINE (cationic liposome formulation), LIPOFECTAMINE 2000 (cationic liposome formulation), LIPOFECTAMINE 3000 (cationic liposome formulation) (Life Technologies)). In various embodiments, the resultant protein expression of the present compositions and methods is about 1.1-fold, or about 1.5-fold, or about 2-fold, or about 5-fold, or about 10-fold, or about 15-fold, or about 20-fold, or about 30-fold, or about 50-fold, or greater than about 50-fold greater than the resultant protein expression of commercially available products (e.g. LIPOFECTIN (cationic liposome formulation), LIPOFECTAMINE (cationic liposome formulation), LIPOFECTAMINE 2000 (cationic liposome formulation), LIPOFECTAMINE 3000 (cationic liposome formulation) (Life Technologies)).

(52) In embodiments, the present invention relates to the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles for transfection, or methods of transfection, which allow for transfection, including efficient transfection as described herein, in various cell types. In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers for transfection, or methods of transfection, allow for transfection, including efficient transfection as described herein, in established cell lines, hard-to-transfect cells, primary cells, stem cells, and blood cells. In embodiments, the cell type is a keratinocyte, a fibroblast, a PBMC, or a dendritic cell.

(53) In embodiments, a present compound (e.g. of Formula I), pharmaceutical composition and/or a lipid aggregate and/or a lipid carrier and/or lipid nucleic-acid complex and/or liposome and/or lipid nanoparticle is suitable for transfection or delivery of compounds to target cells, either in vitro or in vivo.

(54) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles for transfection, or methods of transfection, do not require additional reagents for transfection, e.g. PLUS™ Reagent (DNA pre-complexation reagent) (Life Technologies).

(55) In embodiments, the present compounds (e.g. of Formula I) are components of a pharmaceutical composition and/or a lipid aggregate and/or a lipid carrier and/or lipid nucleic-acid complex and/or liposome and/or lipid nanoparticle which does not require an additional or helper lipid, e.g. for efficient transfection. In embodiments, the pharmaceutical composition and/or a lipid aggregate and/or a lipid carrier and/or lipid nucleic-acid complex and/or liposome and/or lipid nanoparticle does not require one or more of: DOPE, DOPC, cholesterol, and a polyethylene glycol (PEG)-modified lipid (inclusive, without limitation, of a PEGylated PE phospholipid, PC phospholipid, and/or cholesterol), e.g. for efficient transfection.

(56) In embodiments, the present compounds (e.g. of Formula I) are components of a pharmaceutical composition and/or a lipid aggregate and/or a lipid carrier and/or lipid nucleic-acid complex and/or liposome and/or lipid nanoparticle that further comprises an additional or helper lipid.

(57) In embodiments, the additional or helper lipid is selected from one or more of the following categories: cationic lipids; anionic lipids; neutral lipids; multi-valent charged lipids; and zwitterionic lipids. In some embodiments, a cationic lipid may be used to facilitate a charge-charge interaction with nucleic acids.

(58) In embodiments, the additional or helper lipid is a neutral lipid. In embodiments, the neutral lipid is dioleoylphosphatidylethanolamine (DOPE), 1,2-Dioleoyl-sn-glycero-3-phosphocholine (DOPC), or cholesterol. In embodiments, cholesterol is derived from plant sources. In other embodiments, cholesterol is derived from animal, fungal, bacterial or archaeal sources.

(59) In embodiments, the additional or helper lipid is a cationic lipid. In embodiments, the cationic lipid is N-[1-(2,3-dioleoyloxy)propyl]-N,N,N-trimethylammonium chloride (DOTMA), 1,2-bis(oleoyloxy)-3-3-(trimethylammonia) propane (DOTAP), or 1,2-dioleoyl-3-dimethylammonium-propane (DODAP).

(60) In embodiments, one or more of the phospholipids 18:0 PC, 18:1 PC, 18:2 PC, DMPC, DSPE, DOPE, 18:2 PE, DMPE, or a combination thereof are used as helper lipids. In embodiments, the additional or helper lipid is DOTMA and DOPE, optionally in a ratio of about 1:1. In embodiments, the additional or helper lipid is DHDOS and DOPE, optionally in a ratio of about 1:1.

(61) In embodiments, the additional or helper lipid is a commercially available product (e.g. LIPOFECTIN (cationic liposome formulation), LIPOFECTAMINE (cationic liposome formulation), LIPOFECTAMINE 2000 (cationic liposome formulation), LIPOFECTAMINE 3000 (cationic liposome formulation) (Life Technologies)).

(62) In embodiments, the additional or helper lipid is a compound having the Formula (A):

(63) ##STR00019##

where, R1 and R4 are straight-chain alkenyl having 17 carbon atoms; R2 and R5 are —(CH2)p-NH2 where p is 1-4; I is 1-10; and Xa is a physiologically acceptable anion.

(64) In one embodiment, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles include one or more polyethylene glycol (PEG) chains, optionally selected from PEG200, PEG300, PEG400, PEG600, PEG800, PEG1000, PEG1500, PEG2000, PEG3000, and PEG4000. In embodiments, the PEG is PEG2000. In embodiments, the

present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles include 1,2-distearoyl-sn-glycero-3-phosphoethanolamine (DSPE) or a derivative thereof. In one embodiment, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles comprise 1,2-distearoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(polyethylene glycol)-2000] (DSPE-PEG); in another embodiment, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles comprise 1,2-dimyristoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(polyethylene glycol)-2000] (DMPE-PEG); in yet another embodiment, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles comprise 1,2-dimyristoyl-rac-glycero-3-methoxypolyethylene glycol-2000 (DMG-PEG). In further embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles comprise a mixture of PEGylated lipids and/or free PEG chains.

(65) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles comprise one or more of N-(carbonyl-ethoxypolyethylene glycol 2000)-1,2-distearoyl-sn-glycero-3-phosphoethanolamine (MPEG2000-DSPE), fully hydrogenated phosphatidylcholine, cholesterol, LIPOFECTAMINE 2000 (cationic liposome formulation), LIPOFECTAMINE 3000 (cationic liposome formulation), a cationic lipid, a polycationic lipid, and 1,2-distearoyl-sn-glycero-3-phosphoethanolamine-N-[folate(polyethylene glycol)-5000] (FA-MPEG5000-DSPE).

(66) In one embodiment, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles comprise about 3.2 mg/mL N-(carbonyl-ethoxypolyethylene glycol 2000)-1,2-distearoyl-sn-glycero-3-phosphoethanolamine (MPEG2000-DSPE), about 9.6 mg/mL fully hydrogenated phosphatidylcholine, about 3.2 mg/mL cholesterol, about 2 mg/mL ammonium sulfate, and histidine as a buffer, with about 0.27 mg/mL 1,2-distearoyl-sn-glycero-3-phosphoethanolamine-N-[folate(polyethylene glycol)-5000] (FA-MPEG5000-DSPE). In another embodiment, the nucleic acids are complexed by combining 1 μL of LIPOFECTAMINE 3000 (cationic liposome formulation) per about 1 μg of nucleic acid and incubating at room temperature for at least about 5 minutes. In one embodiment, the LIPOFECTAMINE 3000 (cationic liposome formulation) is a solution comprising a lipid at a concentration of about 1 mg/mL. In embodiments, nucleic acids are encapsulated by combining about 1 μg, or about 2 μg, or about 5 μg, or about 10 μg of the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles per about 1 μg of nucleic acid and incubating at room temperature for about 5 minutes or longer than about 5 minutes.

(67) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles comprise one or more nanoparticles. In one embodiment, the nanoparticle is a polymeric nanoparticle. In various embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles comprise one or more of a diblock copolymer, a triblock copolymer, a tetrablock copolymer, and a multiblock copolymer. In various embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid

complexes and/or liposomes and/or lipid nanoparticles comprise one or more of polymeric nanoparticles comprising a polyethylene glycol (PEG)-modified polylactic acid (PLA) diblock copolymer (PLA-PEG), PEG-polypropylene glycol-PEG-modified PLA-tetrablock copolymer (PLA-PEG-PPG-PEG), and Poly(lactic-co-glycolic acid) copolymer. In another embodiment, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles comprise a statistical, or an alternating, or a periodic copolymer, or any other sort of polymer.

(68) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles comprise one or more lipids that are described in WO/2000/027795, the entire contents of which are hereby incorporated herein by reference.

(69) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles comprises Polybrene™ (hexadimethrine bromide) as described in U.S. Pat. No. 5,627,159, the entire contents of which are incorporated herein by reference.

(70) In various embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles comprise one or more polymers. Examples of polymer include hexadimethrine bromide (Polybrene™), DEAE-Dextran, protamine, protamine sulfate, poly-L-lysine, or poly-D-lysine. These polymers may be used in combination with cationic lipids to result in synergistic effects on uptake by cells, stability of the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles, including serum stability (e.g., stability in vivo), endosomal escape, cell viability, and protein expression.

(71) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles are suitable for associating with a nucleic acid, inclusive of, for instance, include any oligonucleotide or polynucleotide.

(72) In embodiments, nucleic acids are fully encapsulated within the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles. In other embodiments, nucleic acids are partially encapsulated within the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles. In still other embodiments, nucleic acids and the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles are both present with no encapsulation of the nucleic acids within the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles.

(73) Fully encapsulated can indicate that the nucleic acid in the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles is not significantly degraded after exposure to serum or a nuclease assay that would significantly degrade free nucleic acids. In embodiments, less than about 25% of particle nucleic acid is degraded in a treatment that would normally degrade about 100% of free nucleic acid. In embodiments, less than about 10% or less than about 5% of the particle nucleic acid is degraded.

(74) Extent of encapsulation may be determined by an Oligreen assay. Oligreen is an ultra-sensitive fluorescent nucleic acid stain for quantitating oligonucleotides and single-stranded DNA in solution

(available from Invitrogen Corporation, Carlsbad, CA).
(75) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles are serum stable and/or they do not rapidly decompose into their component parts upon in vivo administration.
(76) In embodiments, the present compounds (e.g. of Formula I) and/or pharmaceutical compositions and/or lipid aggregates and/or lipid carriers and/or lipid nucleic-acid complexes and/or liposomes and/or lipid nanoparticles are complexed with a nucleic acid (e.g. DNA or RNA) in a ratio, which may depend on the target cell type, generally ranging from about 1:16 to about 25:1 ng lipid:ng DNA or RNA. Illustrative lipid:DNA or RNA ratios are from about 1:1 to about 10:1, e.g. about 1:1, or about 2:1, or about 3:1, or about 4:1 or about 5:1, or about 6:1, or about 7:1, or about 8:1, or about 9:1, or about 10:1.
(77) In embodiments, additional parameters such as nucleic acid concentration, buffer type and concentration, etc., are selected to achieve a desired transfection efficiency, e.g., high transfection efficiency.
(78) In embodiments, the nucleic acid is selected from RNA or DNA.
(79) In embodiments, the DNA is a plasmid, cosmid, phage, recombinant virus or other vector. In embodiments, a vector (or plasmid) refers to discrete elements that are used to, for example, introduce heterologous nucleic acid into cells for expression or replication thereof. In embodiments, the vectors can remain episomal or can be designed to effect integration of a gene or portion thereof into a chromosome of the genome. Also contemplated are vectors that are artificial chromosomes, such as yeast artificial chromosomes and mammalian artificial chromosomes. Included, without limitation, are vectors capable of expressing DNA that is operatively linked with regulatory sequences, such as promoter regions, that are capable of effecting expression of such DNA fragments (e.g. expression vectors). Thus, a vector can refer to a recombinant DNA or RNA construct, such as a plasmid, a phage, recombinant virus or other vector that, upon introduction into an appropriate host cell, results in expression of the DNA. Appropriate vectors can include, without limitation, those that are replicable in eukaryotic cells and/or prokaryotic cells and those that remain episomal or those that integrate into the host cell genome.
(80) In embodiments, the nucleic acid is an RNA, messenger RNA (mRNA), a small interfering RNA (siRNA), micro RNA (miRNA), long non-coding RNA (lncRNA), antisense oligonucleotide, ribozyme, plasmid, immune stimulating nucleic acid, antisense, antagomir, antimir, microRNA mimic, supermir, U1 adaptor, or aptamer.
(81) In embodiments, the RNA is a synthetic RNA. In embodiments, the RNA is a chemically synthesized RNA. In embodiments, the RNA is an in vitro transcribed RNA.
(82) In embodiments, the synthetic RNA (inclusive, without limitation, of mRNA) does not comprise a non-canonical nucleotide. In embodiments, the synthetic RNA (inclusive, without limitation of mRNA) comprises one or more non-canonical nucleotides. In embodiments, the one or more non-canonical nucleotides is selected from 2-thiouridine, 5-azauridine, pseudouridine, 4-thiouridine, 5-methyluridine, 5-methylpseudouridine, 5-aminouridine, 5-aminopseudouridine, 5-hydroxyuridine, 5-hydroxypseudouridine, 5-methoxyuridine, 5-methoxypseudouridine, 5-ethoxyuridine, 5-ethoxypseudouridine, 5-hydroxymethyluridine, 5-hydroxymethylpseudouridine, 5-carboxyuridine, 5-carboxypseudouridine, 5-formyluridine, 5-formylpseudouridine, 5-methyl-5-azauridine, 5-amino-5-azauridine, 5-hydroxy-5-azauridine, 5-methylpseudouridine, 5-aminopseudouridine, 5-hydroxypseudouridine, 4-thio-5-azauridine, 4-thiopseudouridine, 4-thio-5-methyluridine, 4-thio-5-aminouridine, 4-thio-5-hydroxyuridine, 4-thio-5-methyl-5-azauridine, 4-thio-5-amino-5-azauridine, 4-thio-5-hydroxy-5-azauridine, 4-thio-5-methylpseudouridine, 4-thio-5-aminopseudouridine, 4-thio-5-hydroxypseudouridine, 2-thiocytidine, 5-azacytidine, pseudoisocytidine, N4-methylcytidine, N4-aminocytidine, N4-hydroxycytidine, 5-methylcytidine, 5-aminocytidine, 5-hydroxycytidine, 5-methoxycytidine, 5-ethoxycytidine, 5-

hydroxymethylcytidine, 5-carboxycytidine, 5-formylcytydine, 5-methyl-5-azacytidine, 5-amino-5-azacytidine, 5-hydroxy-5-azacytidine, 5-methylpseudoisocytidine, 5-aminopseudoisocytidine, 5-hydroxypseudoisocytidine, N4-methyl-5-azacytidine, N4-methylpseudoisocytidine, 2-thio-5-azacytidine, 2-thiopseudoisocytidine, 2-thio-N4-methylcytidine, 2-thio-N4-aminocytidine, 2-thio-N4-hydroxycytidine, 2-thio-5-methylcytidine, 2-thio-5-aminocytidine, 2-thio-5-hydroxycytidine, 2-thio-5-methyl-5-azacytidine, 2-thio-5-amino-5-azacytidine, 2-thio-5-hydroxy-5-azacytidine, 2-thio-5-methylpseudoisocytidine, 2-thio-5-aminopseudoisocytidine, 2-thio-5-hydroxypseudoisocytidine, 2-thio-N4-methyl-5-azacytidine, 2-thio-N4-methylpseudoisocytidine, N4-methyl-5-methylcytidine, N4-methyl-5-aminocytidine, N4-methyl-5-hydroxycytidine, N4-methyl-5-methyl-5-azacytidine, N4-methyl-5-amino-5-azacytidine, N4-methyl-5-hydroxy-5-azacytidine, N4-methyl-5-methylpseudoisocytidine, N4-methyl-5-aminopseudoisocytidine, N4-methyl-5-hydroxypseudoisocytidine, N4-amino-5-azacytidine, N4-aminopseudoisocytidine, N4-amino-5-methylcytidine, N4-amino-5-aminocytidine, N4-amino-5-hydroxycytidine, N4-amino-5-methyl-5-azacytidine, N4-amino-5-amino-5-azacytidine, N4-amino-5-hydroxy-5-azacytidine, N4-amino-5-methylpseudoisocytidine, N4-amino-5-aminopseudoisocytidine, N4-amino-5-hydroxypseudoisocytidine, N4-hydroxy-5-azacytidine, N4-hydroxypseudoisocytidine, N4-hydroxy-5-methylcytidine, N4-hydroxy-5-aminocytidine, N4-hydroxy-5-hydroxycytidine, N4-hydroxy-5-methyl-5-azacytidine, N4-hydroxy-5-amino-5-azacytidine, N4-hydroxy-5-hydroxy-5-azacytidine, N4-hydroxy-5-methylpseudoisocytidine, N4-hydroxy-5-aminopseudoisocytidine, N4-hydroxy-5-hydroxypseudoisocytidine, 2-thio-N4-methyl-5-methylcytidine, 2-thio-N4-methyl-5-aminocytidine, 2-thio-N4-methyl-5-hydroxycytidine, 2-thio-N4-methyl-5-methyl-5-azacytidine, 2-thio-N4-methyl-5-amino-5-azacytidine, 2-thio-N4-methyl-5-hydroxy-5-azacytidine, 2-thio-N4-methyl-5-methylpseudoisocytidine, 2-thio-N4-methyl-5-aminopseudoisocytidine, 2-thio-N4-methyl-5-hydroxypseudoisocytidine, 2-thio-N4-amino-5-azacytidine, 2-thio-N4-aminopseudoisocytidine, 2-thio-N4-amino-5-methylcytidine, 2-thio-N4-amino-5-aminocytidine, 2-thio-N4-amino-5-hydroxycytidine, 2-thio-N4-amino-5-methyl-5-azacytidine, 2-thio-N4-amino-5-amino-5-azacytidine, 2-thio-N4-amino-5-hydroxy-5-azacytidine, 2-thio-N4-amino-5-methylpseudoisocytidine, 2-thio-N4-amino-5-aminopseudoisocytidine, 2-thio-N4-amino-5-hydroxypseudoisocytidine, 2-thio-N4-hydroxy-5-azacytidine, 2-thio-N4-hydroxypseudoisocytidine, 2-thio-N4-hydroxy-5-methylcytidine, N4-hydroxy-5-aminocytidine, 2-thio-N4-hydroxy-5-hydroxycytidine, 2-thio-N4-hydroxy-5-methyl-5-azacytidine, 2-thio-N4-hydroxy-5-amino-5-azacytidine, 2-thio-N4-hydroxy-5-hydroxy-5-azacytidine, 2-thio-N4-hydroxy-5-methylpseudoisocytidine, 2-thio-N4-hydroxy-5-aminopseudoisocytidine, 2-thio-N4-hydroxy-5-hydroxypseudoisocytidine, N6-methyladenosine, N6-aminoadenosine, N6-hydroxyadenosine, 7-deazaadenosine, 8-azaadenosine, N6-methyl-7-deazaadenosine, N6-methyl-8-azaadenosine, 7-deaza-8-azaadenosine, N6-methyl-7-deaza-8-azaadenosine, N6-amino-7-deazaadenosine, N6-amino-8-azaadenosine, N6-amino-7-deaza-8-azaadenosine, N6-hydroxyadenosine, N6-hydroxy-7-deazaadenosine, N6-hydroxy-8-azaadenosine, N6-hydroxy-7-deaza-8-azaadenosine, 6-thioguanosine, 7-deazaguanosine, 8-azaguanosine, 6-thio-7-deazaguanosine, 6-thio-8-azaguanosine, 7-deaza-8-azaguanosine, and 6-thio-7-deaza-8-azaguanosine.

(83) In embodiments, the compound, pharmaceutical composition, or lipid aggregate described herein is complexed with or associates with a nucleic acid (e.g. DNA or RNA, e.g. mRNA) and the nucleic acid encodes a protein of interest. In embodiments, the protein of interest is a soluble protein. In embodiments, the protein of interest is one or more of a reprogramming protein and a gene-editing protein.

(84) This invention is further illustrated by the following non-limiting examples.

EXAMPLES

Example 1: Synthesis of Linoleoyl Chloride (1)

(85) ##STR00020##

(86) Oxalyl chloride (47.0 mL, 555 mmol) was added to a solution of linoleic acid (70.0 g, 250

mmol) in 580 mL anhydrous methylene chloride at 0° C. under N.sub.2 atmosphere. The reaction was warmed to room temperature and stirred vigorously for 24 hours. Solvent and oxalyl chloride were removed under reduced pressure to yield linoleoyl chloride as a brown oil, which was used without further purification.

Example 2: Synthesis of N1,N4-dilinoleoyl-diaminobutane (2)

(87) A solution of 1,4-diaminobutane (0.428 g, 4.86 mmol) and triethylamine (2.03 mL, 14.6 mmol) in 1 mL of anhydrous methylene chloride was slowly added to a solution of linoleoyl chloride (2.98 g, 10.0 mmol) in 30 mL of anhydrous methylene chloride in an ice bath at 0° C. The reaction mixture was stirred vigorously with a magnetic stir bar. After addition was complete, the ice bath was removed and the mixture was stirred at room temperature for 2.5 days. The reaction was cooled to 4° C., and a white solid precipitated from the solution. The excess linoleoyl chloride was removed by vacuum filtration. The precipitate was washed twice with 10 mL of methylene chloride. The mother liquor was concentrated and more product precipitated. This precipitate was filtered and combined with the previous precipitate. The resulting solid was vacuum dried for 4 hours. A total of 1.9 g of a white solid of the desired product, N.sup.1,N.sup.4-dilinoleoyl-diaminobutane, was obtained.

Example 3: Synthesis of N1,N4-dilinoleyl-diaminobutane (3)

(88) Lithium aluminum hydride (0.6 g, 95%, 16 mmol) was carefully added to a suspension of N.sup.1,N.sup.4-dilinoleoyl-diaminobutane (1.8 g, 2.9 mmol) in 50 mL anhydrous diethyl ether at 0° C. After addition was complete, the ice bath was removed. The reaction mixture was warmed slowly to room temperature and then heated gently to reflux with an appropriate condensing device and stirred for 12 hours. The reaction mixture was cooled and quenched carefully at 0° C. with 5 mL of water. The diethyl ether was removed under reduced pressure, and the reaction mixture was dried under vacuum. The dried reaction mixture was extracted three times with 25 mL of isopropyl alcohol at 80° C. The isopropyl alcohol was removed to yield 1.6 g of oily colorless N.sup.1,N.sup.4-dilinoleyl-diaminobutane.

Example 4: Synthesis of N1,N4-dilinoleyl-N1,N4-di-[2-hydroxy-3-(N-phthalamido)propyl]-diaminobutane (4)

(89) Diisopropylethylamine (1.15 mL, 12.0 mmol) was added to a suspension of N.sup.1,N.sup.4-dilinoleyl-diaminobutane (1.6 g, 2.7 mmol) and N-(2,3-epoxypropyl)-phthalimide (1.6 g, 7.9 mmol) in 12 mL of dry N,N-dimethylformamide. After purging with nitrogen, the reaction mixture was sealed in a round-bottom flask and heated to around 90° C. for 24 hours. N,N-dimethylformamide and diisopropylethylamine were removed and a yellow oil was obtained. Synthesis was continued without additional purification.

Example 5: Synthesis of N1,N4-dilinoleyl-N1,N4-di-(2-hydroxy-3-aminopropyl)-diaminobutane (5)

(90) The entire crude oil of N.sup.1,N.sup.4-dilinoleyl-N.sup.1,N.sup.4-di-[2-hydroxy-3-(N-phthalamido)propyl]-diaminobutane was dissolved in 25 mL of anhydrous ethanol. Hydrazine (0.5 mL, 64-65% aq., 10.3 mmol) was added at room temperature. With an appropriate condensing device, the reaction mixture was heated to reflux. The oil bath was set to 85° C. After 15 minutes, a white solid precipitated from the solution. The reaction mixture was stirred at reflux for 4 hours before being cooled to −20° C. The white solid was removed by gravity filtration. The residue was washed twice with cold ethanol. The combined ethanol solution was concentrated and dried overnight under vacuum. The crude product was extracted with acetone. The combined acetone solution was concentrated and dried overnight under vacuum. 1.0 g of an oil, N.sup.1,N.sup.4-dilinoleyl-N.sup.1,N.sup.4-di-(2-hydroxy-3-aminopropyl)-diaminobutane (referred to herein as DHDLinS, see FIG. **9**B), was obtained. The proton NMR spectrum of a 1-2 mg sample of DHDLinS in 0.6 mL deuterated chloroform was measured on a 500 MHz Varian Inova instrument (FIG. **9**A).

Example 6: Synthesis of Lipids

(91) The following compounds were synthesized by the methods of Examples 1 through 5 using the corresponding amine: N.sup.1,N.sup.4-dilinolenyl-N.sup.1,N.sup.4-di-(2-hydroxy-3-aminopropyl)-diaminobutane (6); N.sup.1,N.sup.2-dilinoleyl-N.sup.1,N.sup.2-di-(2-hydroxy-3-aminopropyl)-diaminoethane (7); N.sup.1,N.sup.3-dilinoleyl-N.sup.1,N.sup.3-di-(2-hydroxy-3-aminopropyl)-diaminopropane (8); N.sup.1,N.sup.5-dilinoleyl-N.sup.1,N.sup.5-di-(2-hydroxy-3-aminopropyl)-diaminopentane (9); N.sup.1,N.sup.6-dilinoleyl-N.sup.1,N.sup.6-di-(2-hydroxy-3-aminopropyl)-diaminohexane (10); N.sup.1,N.sup.8-dilinoleyl-N.sup.1,N.sup.8-di-(2-hydroxy-3-aminopropyl)-diaminooctane (11); N.sup.1,N.sup.10-dilinoleyl-N.sup.1,N.sup.10-di-(2-hydroxy-3-aminopropyl)-diaminodecane (12); N.sup.1,N.sup.12-dilinoleyl-N,N.sup.12-di-(2-hydroxy-3-aminopropyl)-diaminododecane (13).

Example 7: Transfection with Inventive Lipids

(92) Stock solutions of lipid in ethanol were prepared at concentrations of between 5 mg/mL and 20 mg/mL and stored at −20° C. To perform transfections, nucleic acid was first diluted in DMEM (1 μg of nucleic acid in 50 μL of DMEM), then the desired amount of lipid stock solution was added. After adding the lipid, the solution was mixed thoroughly, and complexes were allowed to form for between about 5 minutes and about 25 minutes before adding to cells. For the experiments depicted in FIG. **1** through FIG. **6**, the lipid was used without the acetone purification described in Example 5.

(93) FIG. **1** depicts transfection of primary human epidermal keratinocytes with in vitro transcribed RNA encoding green fluorescent protein (GFP) complexed with the indicated lipids. As shown in the figure, cells were transfected with high efficiency by DHDLinS.

(94) FIG. **2** depicts a time course of the experiment of FIG. **1**, i.e. florescence measured at the indicated time points following transfection using DHDLinS. A fluorescent signal was detected one hour following transfection, and both the signal intensity and number of fluorescent cells increased for several hours following transfection.

(95) FIG. **3** depicts transfection with various indicated amounts of RNA (in nanograms) and lipid-to-RNA mass ratios (in micrograms of lipid per microgram of RNA). As shown in the figure, all RNA amounts and lipid-to-RNA mass ratios tested yielded a fluorescent signal. In general, larger amounts of RNA yielded a stronger signal and/or larger number of fluorescent cells, while minimal increase in fluorescence signal was observed at lipid-to-RNA mass ratios greater than 5 μg/μg.

(96) FIG. **4** depicts a transfection experiment with human peripheral blood mononuclear cells (hPBMCs) instead of keratinocytes. As shown in the figure, DHDLinS effectively transfected hPBMCs at both lipid-to-RNA mass ratios tested, while no transfection was observed with LIPOFECTAMINE 3000, (cationic liposome formulation, “LF3000”) a commercial transfection reagent.

(97) FIG. **5**A extends the transfection findings to a confluent layer of primary human epidermal keratinocytes instead of hPBMCs. As shown in the figure, DHDLinS effectively transfected confluent primary human epidermal keratinocytes at both lipid-to-RNA mass ratios tested, while the cells treated with LIPOFECTAMINE 3000 (cationic liposome formulation) were not efficiently transfected. FIG. **5**B depicts this at higher magnification. As shown in the figure, DHDLinS effectively transfected confluent primary human epidermal keratinocytes at both lipid-to-RNA mass ratios tested, while the cells treated with LIPOFECTAMINE 3000 (cationic liposome formulation) were not efficiently transfected.

(98) FIG. **6**A depicts the results of an experiment conducted as in FIG. **4**, but with primary human adult dermal fibroblasts instead of hPBMCs. As shown in the figure, DHDLinS effectively transfected primary human adult dermal fibroblasts at both lipid-to-RNA mass ratios tested, while the cells treated with LIPOFECTAMINE 3000 (cationic liposome formulation) were not efficiently transfected. FIG. **6**B depicts this experiment at higher magnification. As shown in the figure, DHDLinS effectively transfected primary human adult dermal fibroblasts at both lipid-to-RNA mass ratios tested, while the cells treated with LIPOFECTAMINE 3000 (cationic liposome

formulation) were not efficiently transfected.
(99) FIG. **7** depicts the results of an experiment conducted as in FIG. **3**, but with DHDLinS purified by extraction with acetone as described in Example 5. As shown in the figure, minimal increase in fluorescence signal was observed at lipid-to-RNA mass ratios greater than 2 μg/μg.
(100) FIG. **8** depicts primary human epidermal keratinocytes cultured in a 24-well plate, and transfected with 100 ng per well of in vitro transcribed RNA encoding green fluorescent protein (GFP) complexed with the compounds of Formula I, where n is as indicated or with LIPOFECTAMINE 2000 (cationic liposome formulation, "L2K") or LIPOFECTAMINE 3000 (cationic liposome formulation, "L3K"). Images were taken eight hours following transfection.
(101) TABLE-US-00001 TABLE 1 Transfection with Inventive Lipids Fluorescence Transfection reagent Intensity Formula I (n = 2) 6623 Formula I (n = 4), a.k.a. "DHDLinS" 8009 Formula I (n = 5) 7554 Formula I (n = 6) 8596 Formula I (n = 8) 9170 Formula I (n = 10) 7842 Formula I (n = 12) 5631 LIPOFECTAMINE 2000 (cationic liposome formulation) 3356 LIPOFECTAMINE 3000 (cationic liposome formulation) 3157
(102) Table 1 depicts the results of an experiment in which 20,000 neonatal human epidermal keratinocytes (HEKn) per well of a 24-well plate were transfected with 100 ng of in vitro transcribed RNA encoding green fluorescent protein (GFP) complexed with the indicated lipids. Fluorescence was measured 24 hours after transfection. Numbers indicate mean fluorescence intensity per cell (a.u.).
EQUIVALENTS
(103) Those skilled in the art will recognize, or be able to ascertain, using no more than routine experimentation, numerous equivalents to the specific embodiments described specifically herein. Such equivalents are intended to be encompassed in the scope of the following claims.
INCORPORATION BY REFERENCE
(104) All patents and publications referenced herein are hereby incorporated by reference in their entireties.

Claims

1. A method for transfecting a cell in vivo with a nucleic acid, comprising contacting the cell with a complex comprising the nucleic acid and a compound of the formula: ##STR00021## wherein the complex is suitable for in vivo delivery.
2. The method of claim 1, wherein the nucleic acid comprises DNA.
3. The method of claim 1, wherein the nucleic acid comprises RNA.
4. The method of claim 1, wherein the nucleic acid is selected from an mRNA, an in vitro transcribed mRNA, an siRNA, a miRNA, an lncRNA, an antisense oligonucleotide, a ribozyme, a plasmid, a cosmid, a phage, a recombinant virus, an episomal vector, an artificial chromosome, a yeast artificial chromosome, a mammalian artificial chromosome, an immune stimulating nucleic acid, an antisense, an antagomir, an antimir, an in vitro transcribed RNA, a microRNA mimic, a supermir, a U1 adaptor, an aptamer, and a synthetic RNA.
5. The method of claim 1, wherein the nucleic acid is an in vitro transcribed mRNA.
6. The method of claim 1, wherein the nucleic acid is an siRNA.
7. The method of claim 1, wherein the nucleic acid is an antisense oligonucleotide.
8. The method of claim 1, wherein the nucleic acid is a plasmid.
9. The method of claim 1, wherein the nucleic acid is an episomal vector.
10. The method of claim 1, wherein the nucleic acid is an immune stimulating nucleic acid.
11. The method of claim 5, wherein the in vitro transcribed mRNA comprises a non-canonical nucleotide.
12. The method of claim 11, wherein the non-canonical nucleotide is selected from 2-thiouridine, 5-azauridine, pseudouridine, 4-thiouridine, 5-methyluridine, 5-methylpseudouridine, 5-

aminouridine, 5-aminopseudouridine, 5-hydroxyuridine, 5-hydroxypseudouridine, 5-methoxyuridine, 5-methoxypseudouridine, 5-ethoxyuridine, 5-ethoxypseudouridine, 5-hydroxymethyluridine, 5-hydroxymethylpseudouridine, 5-carboxyuridine, 5-carboxypseudouridine, 5-formyluridine, 5-formylpseudouridine, 5-methyl-5-azauridine, 5-amino-5-azauridine, 5-hydroxy-5-azauridine, 4-thio-5-azauridine, 4-thiopseudouridine, 4-thio-5-methyluridine, 4-thio-5-aminouridine, 4-thio-5-hydroxyuridine, 4-thio-5-methyl-5-azauridine, 4-thio-5-amino-5-azauridine, 4-thio-5-hydroxy-5-azauridine, 4-thio-5-methylpseudouridine, 4-thio-5-aminopseudouridine, 4-thio-5-hydroxypseudouridine, 2-thiocytidine, 5-azacytidine, pseudoisocytidine, N4-methylcytidine, N4-aminocytidine, N4-hydroxycytidine, 5-methylcytidine, 5-aminocytidine, 5-hydroxycytidine, 5-methoxycytidine, 5-ethoxycytidine, 5-hydroxymethylcytidine, 5-carboxycytidine, 5-formylcytydine, 5-methyl-5-azacytidine, 5-amino-5-azacytidine, 5-hydroxy-5-azacytidine, 5-methylpseudoisocytidine, 5-aminopseudoisocytidine, 5-hydroxypseudoisocytidine, N4-methyl-5-azacytidine, N4-methylpseudoisocytidine, 2-thio-5-azacytidine, 2-thiopseudoisocytidine, 2-thio-N4-methylcytidine, 2-thio-N4-aminocytidine, 2-thio-N4-hydroxycytidine, 2-thio-5-methylcytidine, 2-thio-5-aminocytidine, 2-thio-5-hydroxycytidine, 2-thio-5-methyl-5-azacytidine, 2-thio-5-amino-5-azacytidine, 2-thio-5-hydroxy-5-azacytidine, 2-thio-5-methylpseudoisocytidine, 2-thio-5-aminopseudoisocytidine, 2-thio-5-hydroxypseudoisocytidine, 2-thio-N4-methyl-5-azacytidine, 2-thio-N4-methylpseudoisocytidine, N4-methyl-5-methylcytidine, N4-methyl-5-aminocytidine, N4-methyl-5-hydroxycytidine, N4-methyl-5-methyl-5-azacytidine, N4-methyl-5-amino-5-azacytidine, N4-methyl-5-hydroxy-5-azacytidine, N4-methyl-5-methylpseudoisocytidine, N4-methyl-5-aminopseudoisocytidine, N4-methyl-5-hydroxypseudoisocytidine, N4-amino-5-azacytidine, N4-aminopseudoisocytidine, N4-amino-5-methylcytidine, N4-amino-5-aminocytidine, N4-amino-5-hydroxycytidine, N4-amino-5-methyl-5-azacytidine, N4-amino-5-amino-5-azacytidine, N4-amino-5-hydroxy-5-azacytidine, N4-amino-5-methylpseudoisocytidine, N4-amino-5-aminopseudoisocytidine, N4-amino-5-hydroxypseudoisocytidine, N4-hydroxy-5-azacytidine, N4-hydroxypseudoisocytidine, N4-hydroxy-5-methylcytidine, N4-hydroxy-5-aminocytidine, N4-hydroxy-5-hydroxycytidine, N4-hydroxy-5-methyl-5-azacytidine, N4-hydroxy-5-amino-5-azacytidine, N4-hydroxy-5-hydroxy-5-azacytidine, N4-hydroxy-5-methylpseudoisocytidine, N4-hydroxy-5-aminopseudoisocytidine, N4-hydroxy-5-hydroxypseudoisocytidine, 2-thio-N4-methyl-5-methylcytidine, 2-thio-N4-methyl-5-aminocytidine, 2-thio-N4-methyl-5-hydroxycytidine, 2-thio-N4-methyl-5-methyl-5-azacytidine, 2-thio-N4-methyl-5-amino-5-azacytidine, 2-thio-N4-methyl-5-hydroxy-5-azacytidine, 2-thio-N4-methyl-5-methylpseudoisocytidine, 2-thio-N4-methyl-5-aminopseudoisocytidine, 2-thio-N4-methyl-5-hydroxypseudoisocytidine, 2-thio-N4-amino-5-azacytidine, 2-thio-N4-aminopseudoisocytidine, 2-thio-N4-amino-5-methylcytidine, 2-thio-N4-amino-5-aminocytidine, 2-thio-N4-amino-5-hydroxycytidine, 2-thio-N4-amino-5-methyl-5-azacytidine, 2-thio-N4-amino-5-amino-5-azacytidine, 2-thio-N4-amino-5-hydroxy-5-azacytidine, 2-thio-N4-amino-5-methylpseudoisocytidine, 2-thio-N4-amino-5-aminopseudoisocytidine, 2-thio-N4-amino-5-hydroxypseudoisocytidine, 2-thio-N4-hydroxy-5-azacytidine, 2-thio-N4-hydroxypseudoisocytidine, 2-thio-N4-hydroxy-5-methylcytidine, 2-thio-N4-hydroxy-5-hydroxycytidine, 2-thio-N4-hydroxy-5-methyl-5-azacytidine, 2-thio-N4-hydroxy-5-amino-5-azacytidine, 2-thio-N4-hydroxy-5-hydroxy-5-azacytidine, 2-thio-N4-hydroxy-5-methylpseudoisocytidine, 2-thio-N4-hydroxy-5-aminopseudoisocytidine, 2-thio-N4-hydroxy-5-hydroxypseudoisocytidine, N6-methyladenosine, N6-aminoadenosine, N6-hydroxyadenosine, 7-deazaadenosine, 8-azaadenosine, N6-methyl-7-deazaadenosine, N6-methyl-8-azaadenosine, 7-deaza-8-azaadenosine, N6-methyl-7-deaza-8-azaadenosine, N6-amino-7-deazaadenosine, N6-amino-8-azaadenosine, N6-amino-7-deaza-8-azaadenosine, N6-hydroxyadenosine, N6-hydroxy-7-deazaadenosine, N6-hydroxy-8-azaadenosine, N6-hydroxy-7-deaza-8-azaadenosine, 6-thioguanosine, 7-deazaguanosine, 8-azaguanosine, 6-thio-7-deazaguanosine, 6-thio-8-azaguanosine, 7-deaza-8-azaguanosine, and 6-thio-7-deaza-8-azaguanosine.

13. The method of claim 12, wherein the non-canonical nucleotide is 5-methoxyuridine.

14. The method of claim 1, wherein the nucleic acid and the compound are associated with a lipid aggregate.
15. The method of claim 14, wherein the lipid aggregate comprises a lipid nanoparticle.
16. The method of claim 14, wherein the lipid aggregate comprises a liposome.
17. The method of claim 14, wherein the lipid aggregate comprises a lipid carrier.
18. The method of claim 14, wherein the nucleic acid and the compound are associated with an additional lipid.
19. The method of claim 18, wherein the additional lipid is selected from dioleoylphosphatidylethanolamine (DOPE), 1,2-Dioleoyl-sn-glycero-3-phosphocholine (DOPC), cholesterol, and a polyethylene glycol (PEG)-modified lipid.